



Research Article

Digital exchange attributes and the risk of closure

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ABSTRACT

Presently, there are over 500 cryptocurrency exchanges worldwide and more than 19,700 different cryptocurrencies. Despite the fact that cryptocurrency trading is possible via private peer-to-peer transactions, more than 90% of trading occurs on organised exchanges, which provide convenience and liquidity. However, centralised cryptocurrency exchanges are regarded as high-value targets by criminals and are often victims of cyberattacks. In this paper, we investigate the risk of cryptocurrency exchange closures and develop predictive models to forecast which markets will close down and which ones will remain active using publicly available data. Our models perform well and reach a high level of classification accuracy. Exchange trading volume, the availability of public information on exchange staff, exchange lifetime, and several cybersecurity features are identified as key attributes in predicting exchange closures. Nevertheless, our models do not account for all sources of risk, e.g., potential fraud and mismanagement of client funds committed by the exchanges themselves, and market participants are encouraged to carefully consider where and how they store their digital assets.

1. Introduction

The phenomenon of cryptocurrency emerged in 2008 after Bitcoin was introduced as the first cryptographic currency that transacts on a peer-to-peer network [1]. The Reserve Bank of Australia (RBA) [2] defines cryptocurrencies as digital tokens that facilitate direct payments through an online system without an intermediary such as a bank. Regarding the value of digital currencies, the RBA states that “cryptocurrencies have no legislated or intrinsic value”, implying that cryptocurrency prices are set by the market forces of demand and supply. Currently, there are more than 500 cryptocurrency exchanges worldwide, and over 19,700 different digital currencies with a total market capitalization of USD 901.52 billion¹ are being traded. While the cryptocurrency market is a relatively new market that has attracted the attention of investors, it has also become a target of criminals, hackers, and various scams associated with emerging technologies, see, e.g., Ref. [3]. In this paper, we study the issue of cryptocurrency exchange closures, which occur with a relatively high frequency and undermine the credibility of the market. In particular, we develop predictive models to forecast which markets will face closure and which exchanges will

remain active, and we discuss exchange attributes that may help investors discern between sound and vulnerable exchanges.

There are five key factors that have contributed to the extraordinary growth of cryptocurrencies, namely: i) decentralisation, ii) anonymity, iii) transactions that are irreversible and immutable, iv) security, and v) fast and easy access. Each of these properties has played an important role in the quick adoption of the new technology and warrants a brief explanation. Unlike traditional fiat currencies, cryptocurrencies are decentralised virtual currencies [4]. The nature of decentralisation refers to both the transfer of control and the decision-making regarding the supply of currency. While fiat currency transactions are cleared by a third party, such as a bank, decentralised payments are conducted directly between two counterparties [5]. Furthermore, the issuance of new coins, e.g., Bitcoins, is not under the control of a government authority but is managed by the rules embedded into the open-source code that runs the network. Cryptocurrencies can be stored in online ‘hot wallets’ or offline ‘cold wallets’, which are accessed using a private key [6]. Anyone with access to a digital wallet and its associated private key² is able to send and receive cryptocurrency payments, which provides a degree of anonymity. Cryptocurrency transfers are relatively fast, cheap, and efficient,

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¹ As of June 23, 2022. Source: <https://coinmarketcap.com/>.

² A private key is a randomly generated string that is always mathematically related to a digital wallet address but is impossible to reverse engineer thanks to encryption, thus securing the funds.

especially relative to international transfers of fiat currencies [7]. Finally, once a transaction has been processed, it is recorded on the blockchain and cannot be cancelled or modified, i.e., it becomes irreversible and immutable [8].

Despite the fact that cryptocurrency trading is possible via anonymous peer-to-peer transfers, most of the transactions occur on exchanges that require users to authenticate themselves. In fact, it is estimated that more than 90% of cryptocurrency transactions are made through centralised exchanges [9], which is the topic of this paper. There are two key reasons behind the popularity of centralised exchanges. First, as reported in Ref. [10], centralised exchanges charge low transaction costs and provide liquidity. Second, centralised exchanges offer custodian services whereby their clients can store cryptocurrencies in digital wallets managed by their exchange. Such facilities eliminate the need for clients to set up their own digital wallets, which in some cases can be a technically challenging process.

As cryptocurrencies become a more mainstream form of investment and finance, many market participants remain unaware of the risks present in the cryptocurrency markets. One of the main sources of risk is the widespread practise of storing cryptocurrencies on digital exchanges, as discussed above. The lack of regulation in the cryptocurrency market, the amount of money stored in exchange wallets, and the exuberant growth of cryptocurrency values have led to cryptocurrency exchanges becoming high-value targets of criminals, see, e.g., Ref. [11]. Indeed, digital exchanges exhibit a relatively high incidence of cyberattacks and closures that result in the loss of investor funds. In addition to such direct costs, there are also indirect effects whereby cryptocurrency prices and volatility are impacted by cryptocurrency exchange cybersecurity breaches, e.g., Refs. [12–15]. Furthermore, cyberattacks that involve the loss of personal financial information have large reputation costs that far exceed the out-of-pocket costs associated with the cyberattacks themselves, as discussed in Ref. [16].

In this paper, we address the risk that investors face from the closures of digital exchanges by attempting to forecast such closures based on publicly available data. Of particular interest is utilising the data on cybersecurity features implemented by the exchanges and the incidence of any previous cyberattacks in the predictions. If it is possible to accurately predict which markets will remain open and which ones will go out of business, then investors can account for this information and avoid exchanges that are likely to face closure. Of course, it is impossible to consider all sources of risk using publicly available information. Of particular concern is the risk of fraud committed by the exchanges themselves. For instance, the recent collapse of the FTX exchange, which secretly used customer funds to bail out its sister company, Alameda Research, would not have been possible to predict using publicly available information. Indeed, the full extent of the intricate web of commercial links between various financial entities involved in the bankruptcy of FTX is only being uncovered after the collapse. Thus, while in this paper we attempt to predict exchange closures resulting from a number of different factors, we do not take into consideration the risk of closures due to financial fraud and mismanagement of client funds and leave this endeavour for future research.

Our analysis begins by compiling a dataset consisting of ten publicly available attributes for 273 cryptocurrency markets. Amongst the list of relevant predictors are five cybersecurity features, exchange lifetimes, and average trading volumes, as well as an index that measures the degree of anti-money laundering regulation in each home country. The remaining two features capture the availability of publicly accessible information about exchange employees and the implementation of a credit card processing system for purchasing cryptocurrencies. We then train ten machine learning (ML) algorithms to predict which cryptocurrency markets are likely to close and which ones will survive. Our models are ranked based on a number of different performance criteria and evaluated on an independent test dataset. The importance of such predictive modelling is discussed in great detail in Ref. [17]. Finally, we identify key exchange attributes that allow us to discriminate between

the exchanges that face closure and the ones that remain active.

We draw from the current literature to compile a list of potential factors behind exchange closures as well as to inform our empirical model building. However, while there is a large body of literature investigating different types of cyberattacks on blockchain technologies, see, e.g., surveys in Refs. [18,19], a few papers focus specifically on the factors driving cryptocurrency exchange closures. Ref. [20] is a seminal paper that studies the reasons behind digital exchange failures. They reported that trading volume is an important attribute inversely related to closures, while experiencing a previous security breach has no statistically significant impact on exchange closures. Oosthoek and Doerr [11] corroborated this result by uncovering a diminishing pattern in exchange closures that result from cyberattacks over the time period 2011–2019. On the other hand, a longitudinal study of 80 cryptocurrency exchanges [21] reported that experiencing a previous cybersecurity breach increased the probability of closure within the same quarter. More recently, Fantazzini and Calabrese [22] uncovered the following factors as being important in predicting cryptocurrency market closures: public developer team, exchange lifetime, cybersecurity grade, and the number of traded cryptocurrencies. Similarly, Milunovich and Lee [23] reported that exchange lifetime, transacted volume, and cybersecurity measures such as security audit, cold storage, and bug bounty programs rank high in terms of feature importance across multiple algorithms.

In this study, we extend the existing literature in three important directions. First, we collect and analyse a database comprising 273 cryptocurrency exchanges and ten potential predictor variables that cover a number of recent high-profile cyberattacks. Second, we apply ten alternative classification algorithms, including cutting-edge methods such as boosting, ensemble, and deep learning models, that could potentially improve our ability to accurately predict closures. Third, we evaluate the classification performance according to seven different measures, including the model confidence set procedure of Ref. [24], which identifies a set of “superior” forecasting models at a certain confidence level.

Our results are summarised as follows. We report a high degree of classification accuracy in forecasting cryptocurrency market closures using publicly available exchange attributes and machine learning models. Our algorithms generalise well to an independent dataset according to the computed test data accuracy, which ranges between 75.6% and 89.0%, depending on the algorithm used. Thus, while most algorithms perform satisfactorily, these results suggest that classification gains can be made by judiciously selecting between predictive models. The best-performing model, according to both the training and test dataset accuracy, is the random forest algorithm. Alternative performance metrics (other than accuracy) largely confirm the out-performance of this model, and the confidence set procedure of Ref. [24] identifies random forest as the only algorithm in the set of superior forecasting models with 90% confidence. Regarding the usefulness of exchange attributes in generating accurate predictions, trading volume, the availability of public information regarding exchange employees, and exchange lifetime are identified as key predictors by multiple algorithms. Cybersecurity features that play an important role in a subset of models are the security audit and bug bounty programs. Market participants are therefore advised to pay special attention to these attributes when selecting which cryptocurrency exchange to trade on while acknowledging that no exchange is perfectly safe.

The rest of the paper is organised as follows. Section 2 describes our dataset, Section 3 outlines the modelling methodology, empirical results are in Section 4, and Section 5 concludes.

2. Data description

We collect a dataset of cryptocurrency exchange attributes from publicly available media sources and previous studies such as Refs. [11, 21, 23]. Additional information on cybersecurity breach events is obtained from lists compiled by Refs. [25–27]. Our dataset contains 273

samples, which represent cryptocurrency exchanges that have operated during some interval of time within the June 2010–January 2021 period.³

Our target variable is the exchange status, which is a binary variable that we name “active” to signify that class 1 (True) corresponds to an exchange that is active/trading, as opposed to being permanently closed in the case of class 0 (False). It is defined more formally in Eq. (1). Amongst the list of exchange attributes are the following categorical attributes, the first five of which represent available cybersecurity attributes:

1. Breach: a variable denoting if an exchange has suffered a previous security breach;
2. Cold-storage: implementation of cryptocurrency wallets that are kept offline;
3. Two-factor: implementation of two-factor authentication;
4. Security-audit: implementation of an independent security audit;
5. Bug-bounty: existence of a bug-bounty program that rewards discovery and reporting of software bugs;
6. Team: availability of publicly accessible information about exchange employees. This variable is similar to the “Team” variable provided by CoinGecko that is found to be important in previous literature, see Ref. [22], but is unavailable for a large number of exchanges in our sample. We compute “team” as follows: 0 if no publicly available information is found on any staff members via internet searches, 0.5 if there is public information on exchange employees or directors but not on both, and 1 if there is publicly accessible information about both exchange employees and directors;
7. Credit-card: implementation of a credit card processing system for buying cryptocurrencies.

Fig. 1 depicts the empirical distribution of the target variable “active” and the seven features listed above. The frequencies of the target variables are similar, with 141 active and 132 permanently closed markets. Thus, about 52% of the 273 exchanges in the dataset are classified as being active. Considering the distribution of previous cyberbreaches, there are 74 instances (27%) of the exchanges that have suffered some form of a security breach, while the remaining 199 markets have not. Two-factor authentication and cold storage facilities are provided by a large majority of the exchanges because they are relatively easy to implement. In particular, 244 instances (89%) of the markets use two-factor authentication mechanisms to verify their clients’ identities, while 212 (78%) of the exchanges provide digital wallets that are stored offline. Independent security audits and bug bounty programs are more costly to implement and hence less often encountered. In the second row of Fig. 1, we observe that security audit is present in about 28% of all exchanges considered here, while bug bounty disclosure programs are similarly implemented in 30% of the sample markets.

In addition to the above-listed security attributes, we also collect three numerical attributes for each exchange:

6. Volume: average daily transaction volume in millions of USD;
7. Lifetime: exchange lifetime in days;
8. aml/cft index: anti-money laundering and combating the financing of terrorism index that captures the extent of anti-money laundering regulation in the home country of each exchange.

To summarise the properties of these numeric variables, we plot their empirical frequencies in Fig. 2. Of particular interest are lifetime and volume variables, which convey information about the relative success

and size of each exchange.

Transaction volume (in millions of USD) is right-skewed, with most of the exchanges recording moderate trading volumes and a small number of markets recording exceptionally large trading numbers. Considering the data in more detail, we compute the median daily volume to be USD 9.41 million and the mean value to be USD 273.09 million. It is important to point out that although trading volume conveys information about exchange size, it is not a perfect measure of it due to the possibility of wash trading where exchanges artificially inflate trading volumes to increase brand awareness, see, e.g., Ref. [28]. While alternative measures, such as the exchange liquidity quantified by the frequency of transactions [29], may reduce the issues associated with wash trading, they require high-frequency tick data and are computationally demanding. Given that our dataset contains 273 exchanges, many of which have closed and for which tick data are unavailable, such an analysis is infeasible and beyond the scope of the present paper. On the other hand, if the ranking of transaction volumes is not significantly altered by wash trading, then the volume feature would still be a useful predictor for exchange closures, which is what we report in our results below.

Exchange lifetime appears to follow a bimodal distribution, with peaks at around 1250 days and 3000 days. Furthermore, the median lifetime is computed to be 1326 days, while the minimum and maximum lifetimes are 19 and 3885 days, respectively. Lastly, the distribution of the anti-money laundering and combating of financing of terrorism (aml/cft) exhibits three peaks at around 13, 25, and 35 out of the maximum value of 49, indicating the diversity in the country of origin of the investigated markets. The largest peak, however, is located in the top range of the index, indicating a high degree of regulation in the home countries such as the United Kingdom, several countries in the European Union, Singapore, and the United States.

Next, we discuss the estimated correlation coefficients provided in Fig. 3. As indicated by the prevalence of red shaded cells, most of the analysed variables are positively correlated. Considering the entries presented across the first column, we observe that the target variable active is most strongly correlated with the following attributes: team (correlation of 0.6), lifetime (0.53), credit-card and security-audit (0.45), and cold-storage (0.4). Thus, the extent of the publicly available information regarding exchange employees may help identify relatively safe cryptocurrency markets and help investors discriminate between strong and vulnerable exchanges. There is also a positive association between the length of time an exchange has been trading and its status as remaining active (not closing down). The presence of cybersecurity features and credit card processing facilities is also positively related to the variable active. On the other hand, active exhibits a small negative correlation (−0.07) with the previous incidence of cybersecurity breaches, and similarly with aml/cft variables (−0.10). Thus, it appears that experiencing a previous cybersecurity breach and the extent of financial regulation in home countries appear to have little relationship with the present status of an exchange.

Most cybersecurity features exhibit relatively strong and positive correlations amongst themselves, as expected. For instance, two-factor and cold-storage record a correlation of 0.53, while bug-bounty's correlation with security-audit is 0.35. It appears that the cryptocurrency markets that implement safe cybersecurity policies appear to do so across multiple measures.

Finally, we consider the pairs of features that exhibit relatively small correlation coefficients that indicate a lack of a linear relationship. Overall, the feature breached exhibits little relation with any of the other variables presented in Fig. 3. It appears that hackers do not discriminate between their targets on the basis of exchange size, lifetime, team, country of origin or credit card processing facility. However, two notable correlations here are between breached and two-factor (−0.22) and cold-storage (−0.17). This implies that these two security features are negatively associated with the incidence of cybersecurity breaches, as expected. Small correlations are also recorded between aml/cft and security-audit and between aml/cft and lifetime. We interpret these

³ A list of cryptocurrency exchanges used in this research is available in the online appendix at <https://sites.google.com/view/online-appendix/home>. Additionally, the entire training and test datasets are available from the corresponding author upon reasonable request.

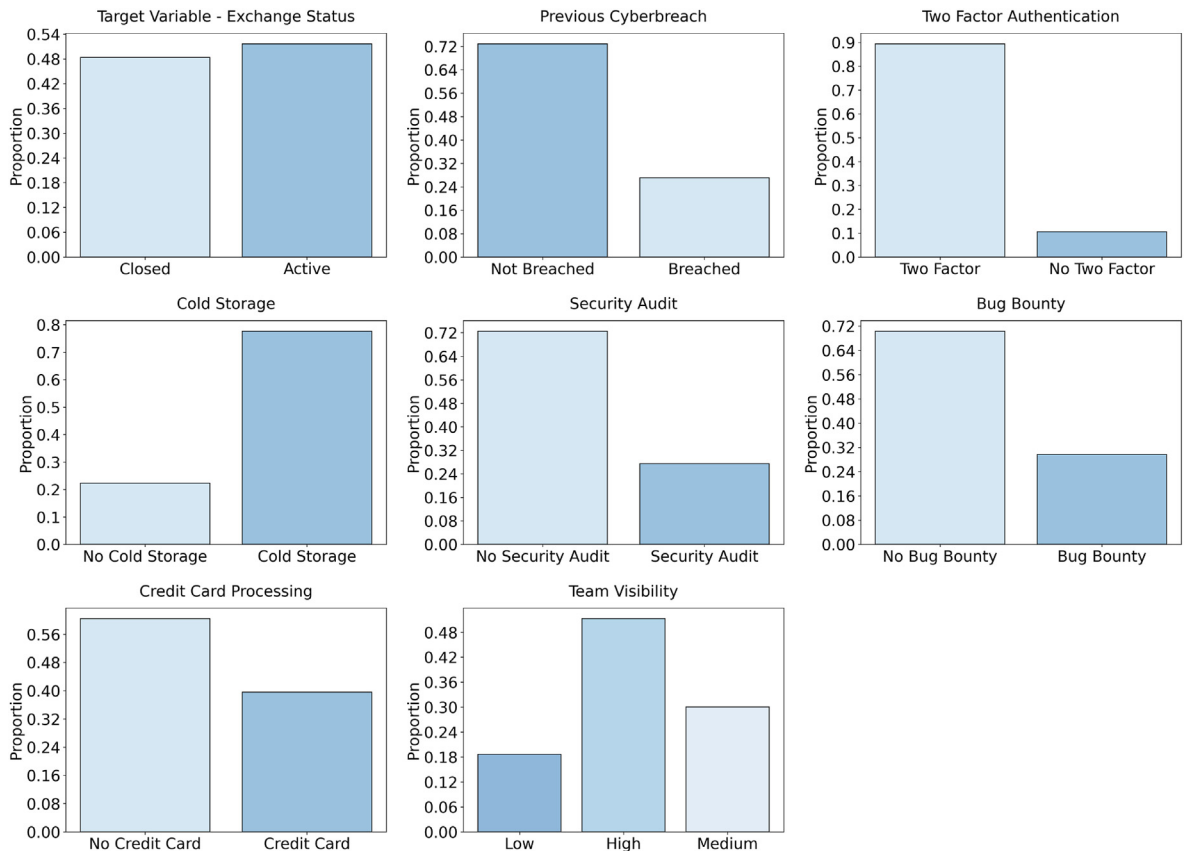


Fig. 1. Empirical frequencies of categorical variables.

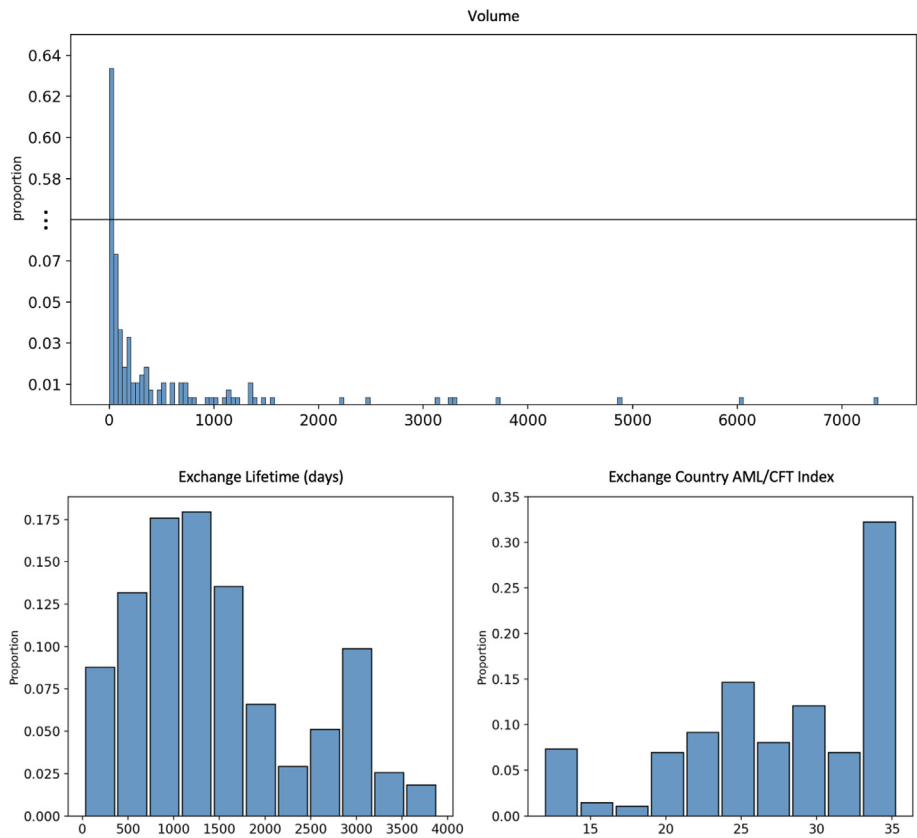


Fig. 2. Empirical frequencies of numerical variables.

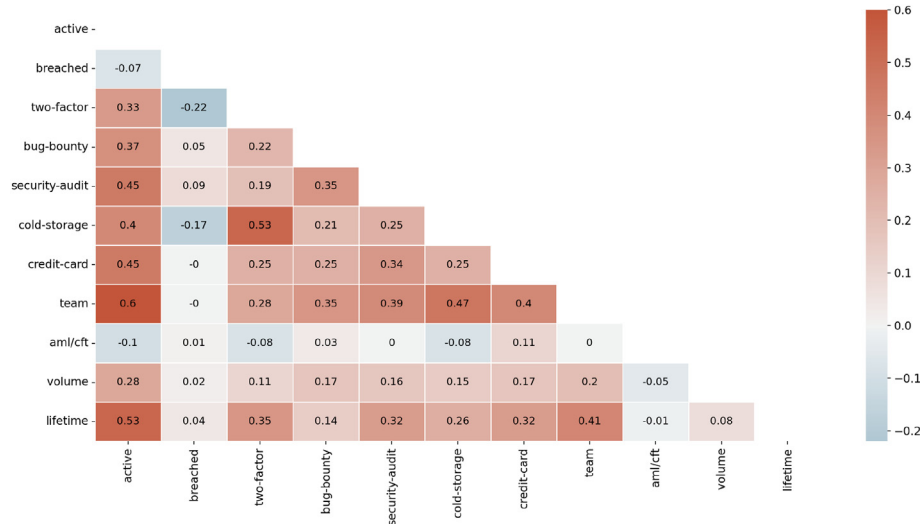


Fig. 3. Correlations heatmap.

correlations to mean that anti-money laundering regulation in the exchange home country has little relationship with the implementation of a security-audit program or with how long the exchange remains operational.

3. Classification methodology

This paper aims to predict which cryptocurrency exchanges will remain active and which will face closure, given a set of exchange attributes. Thus, the target can be defined as a binary variable in the following way

$$y_i = \begin{cases} 1 & \text{digital exchange } i \text{ is active} \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where i enumerates the exchanges in our dataset, i.e., $i = 1, 2, \dots, 273$.

We tackle the forecasting problem by training ten different ML algorithms and selecting the best model according to several measures of classification performance.

3.1. Predictive models

Our list of classification algorithms includes the following models: i) Logistic Regression, ii) Decision Tree, iii) Random Forest, iv) Support Vector Machine, v) Multi-layer Perceptron, vi) KNeighbors, vii) Naive Bayes, viii) AdaBoost, ix) ExtraTrees, and x) Equally-weighted ensemble of the previous nine methods. These models are flexible, and some of them are capable of capturing complicated nonlinear relationships. Below, we provide a brief description of each of these ten classifiers.

Logistic regression was introduced in Ref. [30] and is one of the earliest and most popular classification methods. It specifies the conditional probability of success as $P(y_i = 1|x_i) = \frac{1}{1+e^{-w \cdot x_i}}$ where x_i is the vector of features and w represents the vector of weights that are estimated from the data. An observation is predicted to belong to class 1 (success) if the probability of success is equal to or exceeds 0.5.

A decision tree classifier divides the feature space into rectangles by starting at the tree root and splitting the data on the feature that results in the largest information gain (IG). This procedure is repeated at each node until the leaves are either pure or a limit for maximal depth is reached. We use cross-validation to optimally select tree depth and the criterion used in the computation of IG.

A random forest classifier combines the predictions of multiple decision trees by averaging across their estimated probabilities, with the purpose of reducing the extent of overfitting. A random forest of k trees

may be constructed by drawing a random sample of size n from the training dataset (with replacement) and fitting a decision tree on the bootstrap sample and randomly selected d features. This process is then repeated k times. Finally, all decision trees are combined by averaging their probabilistic predictions and assigning class labels according to the greatest probability. In our application, we cross-validate the optimal number of trees k .

An ExtraTrees (extremely randomised trees) classifier is similar to a random forest, but as its name suggests, the degree of randomness employed in splitting the data is increased. As in random forests, a random subset of candidate features is used, but instead of looking for the most discriminative thresholds, thresholds are drawn at random for each candidate feature, and the best of these randomly generated thresholds is picked as the splitting rule. Here, we estimate the number of base estimators used in ExtraTrees.

Boosting models also combine forecasts of multiple decision trees but put more emphasis on observations that are difficult to classify. AdaBoost classifiers begin by fitting a base classifier on the original dataset and then train additional copies of the classifier on the same dataset but with increased weights for incorrectly classified instances. This makes subsequent classifiers focus on hard-to-classify samples. We use decision tree stumps, i.e., one-level decision trees, as base classifiers and optimise the learning rate and the number of base estimators via cross-validation.

A support vector classifier is designed to be robust to outliers by maximising the margin, which is determined by the distance between the decision boundary and support vectors, i.e., observations closest to the decision boundary. We implement the algorithm with L2 regularisation and optimise the regularisation strength parameter. Furthermore, we account for potential nonlinearities using kernel methods and cross-validate the following kernel values {linear, rbf, polynomial, sigmoid}.

A multilayer perceptron is a feed-forward neural network consisting of three or more layers: input layer, hidden layer(s), and output layer, as depicted in Fig. 4, see, e.g., Ref. [31]. The leftmost layer, known as the input layer, consists of a set of neurons representing the input features. Each neuron in the hidden layer transforms the values from the previous layer with a weighted linear summation $w_1x_1 + w_2x_2 + \dots + w_nx_n$, which is followed by a non-linear activation function $g(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$, for instance, the hyperbolic tan function. The output layer receives the values from the last hidden layer and transforms them into output values. Thus, the units in the hidden layer are fully connected to the input layer, and the output layer is fully connected to the hidden layer. If such a network has more than one hidden unit, it is called a deep artificial neural network (NN). We employ a Leave-One-Out-Cross-Validation (LOOCV) cross-validation to find the optimal values for the number of hidden layers, choosing

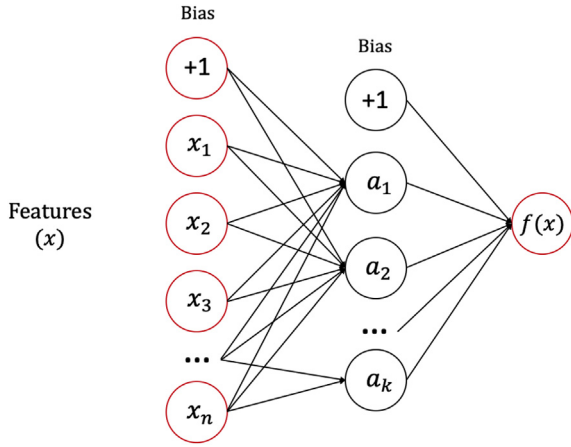


Fig. 4. A hypothetical multilayer perceptron.

between 1 and 2 layers, thus allowing for deep learning. The number of neurons in the hidden layers is set to 8, representing the number of features in the dataset. In addition, we also optimise the strength of L2 regularisation and the activation function, which is selected from the following set {identity, logistic, tanh, and relu}.

The next method we discuss is KNeighbours (K nearest neighbours), which finds a predefined number of training samples closest to the new point and predicts the label from these. The algorithm works by first setting the number of neighbours (k) and a distance metric, and then finding the k nearest neighbours of the data example we need to classify. The classification is completed by assigning the class label by the majority vote of the identified neighbours.

Naive Bayes applies Bayes' theorem with the assumption of conditional independence between each pair of features given the value of the class variable, i.e., $P(x_i|y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = P(x_i|y)$. Substituting this assumption into Bayes' theorem

$$P(y|x_1, \dots, x_n) = \frac{P(y)P(x_1, \dots, x_n|y)}{P(x_1, \dots, x_n)} \quad (2)$$

we obtain

$$P(y|x_1, \dots, x_n) = \frac{P(y) \prod_{i=1}^n P(x_i|y)}{P(x_1, \dots, x_n)}.$$

Finally, since $P(x_1, \dots, x_n)$ is constant, the classification rule may be expressed as follows:

$$\hat{y} = \underset{y}{\operatorname{argmax}} P(y) \prod_{i=1}^n P(x_i|y)$$

where $P(y)$ and $P(x_i|y)$ can be estimated using the Maximum A Posteriori (MAP) method. In this application, we use a Gaussian distribution to model $P(x_i|y)$ and optimise the variance smoothing parameter, which is the portion of the largest variance of all features that are added to variances for calculation stability.

The last model applied in this study is a majority voting ensemble that combines the predictions of the nine previously discussed classifiers.

3.2. Training algorithms and assessing classification ability

Correct evaluation of classification performance requires an independent dataset that has not been used for the purpose of training the classifiers. This is important because machine learning models can be overfitted to the training dataset even when best modelling practises are followed. This can happen, for instance, when there is an insufficient amount of training data. Thus, a portion of the sample data is usually set aside before the analysis and only used for the purpose of out-of-sample

performance evaluation. We apply this method by splitting our database randomly and allocating 70% of the observations (191 observations) to the training dataset and the remaining 30% (82 observations) to the test set. While this training/test ratio is often found in the forecasting literature, we also try alternative 80/20 and 60/40 ratios and find them to produce results similar to the ones reported here.

When fitting the models on the training dataset, we also optimise certain hyperparameters, as described in Section 3.1, using LOOCV grid search. Leave-one-out cross-validation is a special case of cross-validation where the number of folds equals the number of instances k in the training dataset (191 in our case). Thus, $k-1$ folds are used for model training, and 1 fold is kept for performance evaluation. This is repeated k times until all k folds are used for evaluation, and the average performance is computed across all k folds. Once the best hyperparameter values are found via grid search on the basis of the average classification performance, the model is retrained on the complete training set, and the final classification performance measures can be obtained for the training dataset (Table 1), as well as for the independent test dataset (Table 2).

We calculate in-sample (training dataset) and out-of-sample (test dataset) classification performance according to the following measures: i) classification accuracy, ii) precision, iii) recall, iv) F1 Score, v) AUC, vi) Brier Score, and vii) model confidence set. These quantities evaluate different aspects of classification performance that cryptocurrency investors may care about that are explained below. In the following discussion, we represent true positives by TP, true negatives by TN, false positives by FP, and false negatives by FN.

- i) Classification Accuracy = $\frac{\text{number of correctly classified examples}}{\text{sample size}} = \frac{TP+TN}{TP+TN+FP+FN}$. Accuracy is the ratio of correctly predicted observations to the sample size, and it is the most frequently used metric of classification performance.
- ii) Recall = $\frac{TP}{TP+FN}$. Recall is computed as the ratio of the number of correctly predicted positive examples to all positive samples. It tells us how many exchanges were predicted to remain active out of all the exchanges that remained active.
- iii) Precision = $\frac{TP}{TP+FP}$. Precision represents the ratio of true positives to all positively labelled (predicted) examples. It answers the question of how many exchanges remain active out of all the exchanges which are predicted to remain active.
- iv) F1 Score = $2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$. F1 Score is a weighted average of precision and recall.
- v) AUC is the area under the receiver operating characteristic curve (ROC). It is a measure of the model's ability to discriminate between active and closed exchanges. The worst AUC is 0.5, and the best AUC is 1.0.
- vi) Brier Score = $\frac{1}{k} \sum_{i=1}^k (\hat{p}_i - o_i)^2$ where $\hat{p}_i$ is the estimated probability of remaining active while o_i is the actual binary outcome for exchange i . Finally, k is the number of forecasting instances [32].
- vii) Model Confidence Set (MCS) [24] procedure is used to identify a set of "superior" forecasting models at a certain confidence level. In our application, we use 90% confidence and compute MSC on the basis of the Brier Score function $(\hat{p}_i - o_i)^2$.

4. Empirical results

In this section, we provide the results of classification performance and identify key predictors that contribute to accurate forecasts.

4.1. Ranking classifiers

In Table 1, we rank the classifiers according to several classification measures computed from the training dataset. First, it is important to note that all ten classifiers achieve satisfactory performance exhibiting accuracy metrics that exceed 0.822. The best-performing, algorithm is

Table 1

Training dataset classification metrics.

Algorithm	Accuracy	Precision	Recall	F1 Score	AUC	Brier Score	MCS
Random Forest	1.000	1.000	1.000	1.000	1.000	0.012	Yes
Multilayer Perceptron	0.974	1.000	0.949	0.974	0.992	0.025	No
Ensemble	0.942	0.968	0.919	0.943	0.989	0.054	No
AdaBoost	0.932	0.939	0.929	0.934	0.984	0.138	No
Support Vector	0.927	0.938	0.919	0.929	0.971	0.056	No
Logistic Regression	0.901	0.900	0.909	0.905	0.963	0.075	No
KNeighbors	0.890	0.964	0.818	0.885	0.970	0.069	No
Naive Bayes	0.890	0.933	0.848	0.889	0.948	0.094	No
Decision Tree	0.880	0.904	0.859	0.881	0.928	0.090	No
ExtraTrees	0.822	0.892	0.747	0.813	0.898	0.167	No

Notes: Computations based on 191 samples. MCS (model confidence set) is computed based on 90% confidence.

Table 2

Test dataset classification metrics.

Algorithm	Accuracy	Precision	Recall	F1 Score	AUC	Brier Score	MCS
Random Forest	0.890	0.902	0.881	0.892	0.951	0.092	Yes
AdaBoost	0.878	0.921	0.833	0.875	0.965	0.151	No
Logistic Regression	0.854	0.841	0.881	0.860	0.915	0.131	No
Naive Bayes	0.841	0.872	0.810	0.840	0.911	0.130	No
Support Vector	0.841	0.854	0.833	0.843	0.887	0.126	No
Ensemble	0.841	0.854	0.833	0.833	0.933	0.110	No
ExtraTrees	0.829	0.938	0.714	0.811	0.865	0.177	No
KNeighbors	0.805	0.842	0.762	0.800	0.877	0.139	No
Multilayer Perceptron	0.793	0.838	0.738	0.785	0.901	0.163	No
Decision Tree	0.756	0.789	0.714	0.750	0.886	0.138	No

Notes: Computations based on 82 samples. MCS (model confidence set) is computed based on 90% confidence.

the random forest classifier, which reaches a training set accuracy of 1.000. While the accuracy of 1.000 may be unusual and could indicate overfitting, further results presented in [Table 2](#), which are based on an independent test dataset, confirm the superior performance of this classifier relative to the other nine models used here. The second-highest ranking model is the multilayer perceptron algorithm with an accuracy of 0.974, while in third place is the ensemble voting classifier that combines the forecasting of the other nine models with an accuracy of 0.942.

Although the above discussion is based on the accuracy metric, the remaining six measures of performance are unanimous in ranking the random forest algorithm as the best of the ten models. In particular, we observe that Precision, Recall, F1 Score, and AUC are all equal to 1.000 for the random forest model. It also records the lowest (best) Brier Score of 0.012 and is the only model included in the model confidence set (MCS) of Ref. [24] with 90% confidence. The performance measures are also largely consistent in their scoring of the other models, although there are some permutations in the ranked positions. It is of interest to note that the worst-performing model (ExtraTrees) is almost 20% below the best model in terms of accuracy and exhibits a Brier Score many times larger than that of random forest. This demonstrates the importance of the careful selection of predictive models.

While the classification performance measures computed on the training dataset provide an indication of the effectiveness of different classifiers to correctly predict exchange closures, the real test of accuracy is only possible on an independent dataset. The reason for this is that in-sample criteria computed from the training data are likely to be inflated for complex models due to overfitting, even when best modelling practises are followed. For this reason, we present the evaluation criteria that are computed using a completely independent test dataset in [Table 2](#). These results provide a good representation of the true classification ability since the test data have not been used for the purpose of training the algorithms or optimising hyperparameter values.

Comparing the results reported in [Table 2](#) with the figures provided in [Table 1](#), we observe a decrease in the classification metrics (with the exception of the Brier Score, which increases but is a negatively-oriented score where lower values are better). This is of course expected since the

models have not been trained on this dataset. Nevertheless, the reported accuracy is still high, ranging between 0.756 and 0.890, indicating that the models generalise well to unseen samples and retain their ability to discern between safe and vulnerable cryptocurrency markets.

The best-performing algorithm listed in the top row of [Table 2](#) is the random forest classifier with an accuracy of 0.890, suggesting that about 89% of all exchanges in the test dataset are classified correctly. This model is also ranked best according to Recall, F1 Score, and Brier Score. Furthermore, it is the only algorithm identified to be in the confidence set of superior forecasting models with 90% confidence according to the MCS procedure. While the Precision and AUC metrics computed for random forest are slightly below those of AdaBoost, the results provided by the test dataset give us confidence in our modelling technique and identify the random forest algorithm as the best model in the first step, i.e., [Table 1](#). AdaBoost is ranked second according to the accuracy of 0.878 and exhibits satisfactory values for several other criteria as well.

Although the best model in [Table 1](#) remains in the top position when evaluated on test data, some algorithms drop significantly in their ranking. For instance, the multilayer perceptron, which is ranked second in [Table 1](#), can now be found in ninth place in [Table 2](#). This is not necessarily an issue with how the model is trained, since we apply good modelling practises such as LOOCV, but is rather related to the fact that certain classifiers require more data than others in order to reduce the amount of overfitting and generalise well to new datasets. This implies that it is difficult to remedy the situation for certain models without obtaining a larger dataset. Since the quantity of our data is limited by the total number of active and closed cryptocurrency exchanges, our strategy is to recommend a model that generalises well to an independent dataset that is available.

Given that the random forest model exhibits good performance both in-sample and out-of-sample according to multiple criteria, we examine its classification results in more detail by considering the confusion matrix in [Fig. 5](#). Out of the total of 82 samples contained in the test dataset, there are 42 exchanges that remain active (class 1), and 40 exchanges have closed down (class 0). As illustrated in the first row of the figure, five of the 42 active exchanges are misclassified as closed. This



Fig. 5. Confusion matrix (test dataset).

corresponds to the recall value of 0.881 presented in Table 2. In contrast, of the 40 exchanges that went out of business, we successfully predict 36, while 4 exchanges are misclassified as remaining active. This results in a true negative rate (specificity) of 0.9. Thus, while we are able to separate the classes with high accuracy, a certain amount of risk still remains when predicting cryptocurrency exchange closures, and cryptocurrency investors need to exercise a high degree of vigilance when deciding where and how to store their digital assets.

4.2. Identifying key predictors

In order to gain further insight into digital exchange closures, we investigate which features contribute most to the reported classification ability. Feature importance analysis refers to assessing the usefulness of each predictor variable in forecasting the target variable. However, there is no single method of evaluating feature importance that can be applied to all algorithms, and in some cases, no measure of feature importance can be computed at all. From the ten classifiers that we implement here, three models have well-defined measures of importance that are readily available. Should multiple algorithms provide evidence to suggest that a certain feature is “important”, then we can have more confidence in the usefulness of that predictor.

Fig. 6 depicts a measure of Gini importance that is obtained from the random forest classifier. Exchange lifetime (measured in days) appears to be the most useful feature for predicting cryptocurrency exchange closures, while volume (in USD) is placed second. These two predictors are

followed by the team variable, which captures the extent of publicly available information on exchange employees, and then by the aml/cft index and security-audit. The remaining attributes exhibit significantly less importance, with the impacts of breached and two-factor (authentication) being marginal. The result regarding two-factor does not necessarily imply that two-factor authentication bears no importance for the cybersecurity of digital markets but is likely to be an artifact of the composition of our dataset, 90% of which implements two-factor authentication.

Next, we consider feature importance according to the decision tree classifier depicted in Fig. 7. We confirm the importance of the top three predictors identified previously by the random forest algorithm, namely volume, team, and lifetime. Volume now exhibits more than three times the importance of team and lifetime, which have roughly equal importance and rank second and third, respectively. The remaining predictors appear to have no role in the classification of exchange closures according to the decision tree algorithm.

The last piece of evidence is provided by the estimates of the logistic regression model. Table 3 reports marginal effects and their p -value that present a different perspective on feature importance to what we discussed above. Marginal effects measure how the predicted probability of a binary outcome varies according to a risk factor. For instance, we can consider how the probability of remaining active changes between an exchange with a bug-bounty program and an exchange without it.

According to the last column of the table, the estimated coefficients on volume, security-audit, lifetime, bug-bounty, aml/cft, and team exhibit small p -value, and are statistically significant at the 5% level. Thus, we again confirm the importance of the volume, lifetime, and team variables discussed above, as well as identify additional features that are statistically significant in this model. All but one (aml/cft) of the effects are positive, implying that they increase the probability of cryptocurrency exchanges remaining active and not facing closure. For instance, we see that implementing security-audit and bug-bounty programs results in 0.157 and 0.098 improvements in the probability of remaining active, respectively. The feature breached that records the incidence of previous cybersecurity breaches is negative but statistically insignificant at the 5% level, meaning that the estimated effect is likely due to random chance. This is consistent with the findings in Refs. [20, 23]. The result for the aml/cft coefficient suggests that having a home country with a high degree of anti-money laundering regulation reduces the probability of remaining active.

In regard to the possible explanations/intuition behind the results presented in Table 3, we make the following suggestions. Transaction volume is an indicator of exchange size and hence of the extent of financial resources that are available to avert closure in times of financial stress. While this explains the positive and statistically significant coefficient on volume, one must keep in mind that this argument is

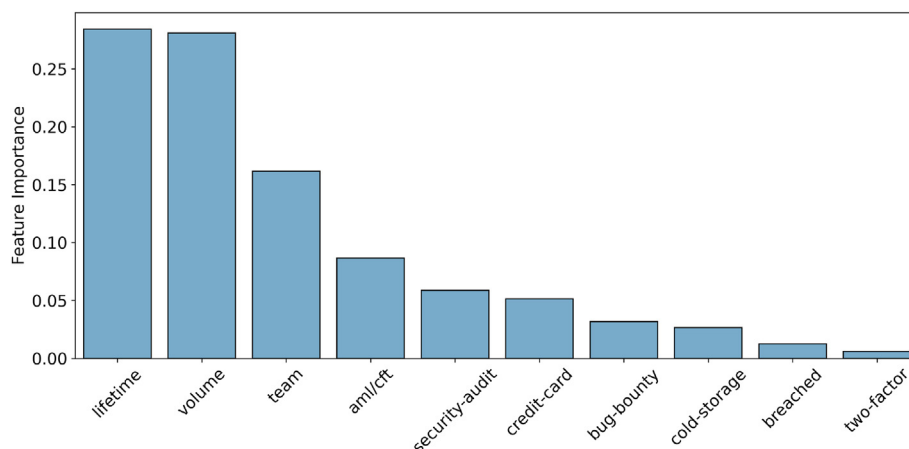


Fig. 6. Random forest feature importance.

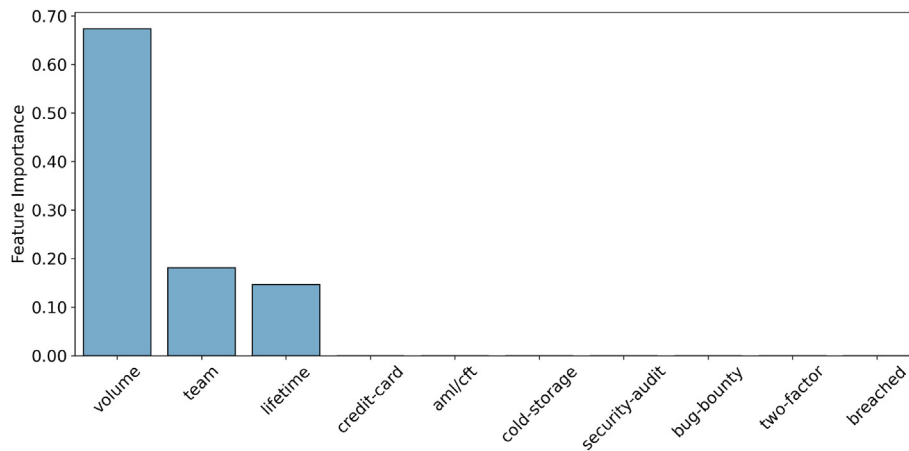


Fig. 7. Decision tree feature importance.

Table 3
Logistic regression marginal effects.

	dy/dx	Std. Err.	z-stat.	p-value
volume	0.206	0.070	2.953	0.003
security-audit	0.157	0.052	3.004	0.003
cold-storage	−0.023	0.060	−0.384	0.701
lifetime	0.154	0.029	5.390	0.000
bug-bounty	0.098	0.048	2.055	0.040
aml/cft	−0.053	0.023	−2.334	0.020
breached	−0.080	0.053	−1.527	0.127
two-factor	0.279	0.159	1.754	0.080
Team	0.097	0.022	4.364	0.000
credit-card	0.033	0.022	1.547	0.122

complicated by the practise of washing trading we discussed in Section 2. A positive coefficient is also expected for the exchange lifetime because longer-running exchanges are likely to have more experienced leadership and expert knowledge. Next, being able to find publicly available information on exchange employees and directors, as reflected in the variable team, provides a level of security against scams that are widespread in the cryptocurrency markets, see, e.g., Ref. [3]. Finally, concerning the cybersecurity features, we note that security audit and bug bounty programs are expected to be statistically significant since they are performed by independent third-party auditors and security engineers, respectively, and provide a signal regarding the security of the exchange that is independent of the exchange itself. On the other hand, cold storage and two-factor authentication can be easily implemented by the markets themselves.

5. Conclusions

Despite the degree of anonymity that peer-to-peer trading provides, over 90% of cryptocurrency transactions are conducted on centralised exchanges that require authentication. This is largely due to the convenience that organised exchanges present, such as low transaction costs and liquidity. Another key facility that centralised exchanges deliver is cryptocurrency accounts, where traders can store digital assets and thus avoid the process of setting up their own cryptocurrency wallets. However, large amounts of cryptocurrency funds stored on exchange platforms attract criminals and result in frequent cyberattacks.

We investigate the risk associated with cryptocurrency exchange closures by attempting to forecast which exchanges will close down and which ones will remain active. In particular, we compile a dataset of ten publicly available exchange attributes on 273 cryptocurrency markets and use these variables in machine learning algorithms to predict exchange closures. Amongst the exchange attributes are traded volumes, exchange lifetime, the availability of public information on exchange

employees, credit card processing facilities, and the exchange home country anti-money laundering regulation index. The list of the predictors is completed by five cybersecurity features, including the variables representing the incidence of previous security breaches for each exchange, as well as the implementation of two-factor authentication, bug bounty, external security, audit and cold storage.

Our analysis suggests that the model best suited for predicting cryptocurrency exchange closures is the random forest classifier. This result is obtained from two independent datasets: a sample of exchanges used for training the models and a different subset of data allocated for testing the classification performance. In particular, the random forest algorithm reaches an out-of-sample (test dataset) classification accuracy of 0.890, which demonstrates its ability to generalise well to new data. Other performance measures such as Precision, Recall, F1, and Brier Scores, as well as AUC, are similarly high and indicate good classification ability. Additionally, out of ten different models considered in our study, random forest is the only algorithm included in the confidence set of superior forecasting models [24]. A possible application of this result can be found in an expert system that provides cryptocurrency exchange rankings driven by probabilistic predictions of remaining active.

In terms of feature importance, we identify trading volume, the availability of public information on exchange employees, and exchange lifetime as key predictors across multiple algorithms. Additionally, cybersecurity attributes such as security audit and bug bounty programs also play an important role in a subset of the investigated models. These attributes can guide investors when choosing a safe cryptocurrency exchange. For instance, an exchange with a long track record of trading, high transaction volume, and publicly accessible information about its staff, directors, and cybersecurity features is preferred to a newly established market with low trading volume, opaque employment structure, and no reference to cybersecurity policies. Finally, it is important to note that since our dataset is based on publicly available information, it does not account for all sources of risk, in particular the risk associated with financial fraud and mismanagement of client funds by the exchanges themselves. Therefore, we encourage investors to investigate different technologies for storing digital assets, such as privately held cryptocurrency wallets, and not only rely on custodian services provided by their digital exchange.

Author Contribution

Seung Ah Lee: Conceptualization, Data curation, Methodology, Formal analysis, Software, Visualization, Writing—original draft, Writing—review & editing. **George Milunovich:** Methodology, Formal analysis, Software, Writing—original draft, Writing—review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] S. Nakamoto, A peer-to-peer electronic cash system. <https://bitcoin.org/bitcoin.pdf>, 2008 (Accessed: 23 Jun 2022).
- [2] RBA, What are cryptocurrencies?. <https://www.rba.gov.au/education/resources/explainers/cryptocurrencies.html>, 2022 (Accessed: 23 Jun 2022).
- [3] P. Xia, H. Wang, B. Zhang, R. Ji, B. Gao, L. Wu, X. Luo, G. Xu, Characterizing cryptocurrency exchange scams, *Comput. Secur.* 98 (2020), 101993.
- [4] L. Poletti, Cryptocurrency basics - 3 key characteristics and why they matter [online]. Available: <https://medium.com/the-capital/key-characteristics-of-cryptocurrency-and-why-do-they-matter-to-you-5f33e483a40f>, 2018 (Accessed: 23 Jun 2022).
- [5] W.J. Luther, S.S. Smith, Is bitcoin a decentralized payment mechanism? *J. Inst. Econ.* 16 (4) (2020) 433–444, <https://doi.org/10.1017/S1744137420000107>.
- [6] M. Guri, Beatcoin: leaking private keys from air-gapped cryptocurrency wallets, in: 2018 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData), IEEE, 2018, pp. 1308–1316, <https://doi.org/10.1109/Cybermatics.2018.2018.00227>.
- [7] J.J. Masilela, R.B. van Wyk, N. Marwa, Assessing the variability of crypto collateral assets in secured lending on the blockchain, *Dev. South Afr.* 39 (3) (2022) 830–840, <https://doi.org/10.1080/0376835X.2021.1906630>.
- [8] E. Feig, A Framework for Blockchain-Based Applications, 2018 arXiv preprint arXiv: 1803.00892.
- [9] N. Roubini, Exploring the cryptocurrency and blockchain ecosystem, *Testimony for the Hearing of the US Senate Committee on Banking, Housing and Community Affairs*, 2018, pp. 115–453.
- [10] A. Barbon, A. Ranaldo, On the Quality of Cryptocurrency Markets: Centralized versus Decentralized Exchanges, 2021 arXiv preprint arXiv:2112.07386.
- [11] K. Oosthoek, C. Doerr, From hodl to heist: analysis of cyber security threats to bitcoin exchanges, in: 2020 IEEE International Conference on Blockchain and Cryptocurrency (ICBC), IEEE, 2020, pp. 1–9, <https://doi.org/10.1109/ICBC48266.2020.9169412>.
- [12] M.S. Brown, B. Douglass, An event study of the effects of cryptocurrency thefts on cryptocurrency prices, in: *Proceedings of the 2020 Spring Simulation Conference (SpringSim)*, IEEE, 2020, pp. 1–12, <https://doi.org/10.22360/SpringSim.2020.CSE.001>.
- [13] J. Hu, Q. Luo, J. Zhang, The fluctuations of bitcoin price during the hacks, *Int. J. Appl. Res. Manag. Econ.* 3 (1) (2020) 10–20, <https://doi.org/10.33422/ijarme.v3i1.278>.
- [14] Š. Lyócsa, P. Molnár, T. Plíhal, M. Širáňová, Impact of macroeconomic news, regulation and hacking exchange markets on the volatility of bitcoin, *J. Econ. Dynam. Control* 119 (2020), 103980, <https://doi.org/10.1016/j.jedc.2020.103980>.
- [15] G. Milunovich, S.A. Lee, Measuring the impact of digital exchange cyberattacks on bitcoin returns, *Econ. Lett.* 221 (2022), 110893, <https://doi.org/10.1016/j.econlet.2022.110893>.
- [16] S. Kamiya, J.-K. Kang, J. Kim, A. Milidonis, R.M. Stulz, Risk management, firm reputation, and the impact of successful cyberattacks on target firms, *J. Financ. Econ.* 139 (3) (2021) 719–749, <https://doi.org/10.1016/j.jfineco.2019.05.019>.
- [17] G. Shmueli, O.R. Koppius, Predictive analytics in information systems research, *MIS Q.* 35 (3) (2011) 553–572, <https://doi.org/10.2307/23042796>.
- [18] S. Ramos, F. Pianese, T. Leach, E. Oliveras, A great disturbance in the crypto: understanding cryptocurrency returns under attacks, *Blockchain: Res. Appl.* 2 (3) (2021), 100021, <https://doi.org/10.1016/j.bcr.2021.100021>.
- [19] H. Hasanova, U.-j. Baek, M.-g. Shin, K. Cho, M.-S. Kim, A survey on blockchain cybersecurity vulnerabilities and possible countermeasures, *Int. J. Netw. Manag.* 29 (2) (2019), e2060, <https://doi.org/10.1002/nem.2060>.
- [20] T. Moore, N. Christin, Beware the middleman: empirical analysis of bitcoin-exchange risk, in: AR. Sadeghi (Eds.), *Financial Cryptography and Data Security*, Springer, Berlin, Heidelberg, 2013, pp. 25–33, https://doi.org/10.1007/978-3-642-39884-1_3.
- [21] T. Moore, N. Christin, J. Szurdi, Revisiting the risks of bitcoin currency exchange closure, *ACM Trans. Internet Technol.* 18 (4) (2018) 1–18, <https://doi.org/10.1145/3155808>.
- [22] D. Fantazzini, R. Calabrese, Crypto exchanges and credit risk: modeling and forecasting the probability of closure, *J. Risk Financ. Manag.* 14 (11) (2021) 516, <https://doi.org/10.3390/jrfm14110516>.
- [23] G. Milunovich, S.A. Lee, Cryptocurrency exchanges: predicting which markets will remain active, *J. Forecast.* 41 (5) (2022) 945–955, <https://doi.org/10.1002/for.2846>.
- [24] P.R. Hansen, A. Lunde, J.M. Nason, The model confidence set, *Econometrica* 79 (2) (2011) 453–497, <https://doi.org/10.3982/ECTA5771>.
- [25] Hackernews, A huge list of cryptocurrency thefts. <https://hackernoon.com/a-huge-list-of-cryptocurrency-thefts-16d6bf246389>, 2019 (Accessed: 23 Jun 2022).
- [26] Selfkey, Comprehensive list of cryptocurrency exchange hacks. <https://selfkey.org/list-of-cryptocurrency-exchange-hacks/>, 2019 (Accessed: 23 Jun 2022).
- [27] Slowmist, Exchange : 94 hack event(s). <https://hacked.slowmist.io/en/?c=Exchange>, 2021 (Accessed: 23 Jun 2022).
- [28] L.W. Cong, X. Li, K. Tang, Y. Yang, Crypto Wash Trading, 2021. SSRN preprint, <https://doi.org/10.2139/ssrn.3530220>.
- [29] J. Kwapien, M. Watorek, M. Bezbradica, M. Crane, T. Tan Mai, S. Drozd, Analysis of inter-transaction time fluctuations in the cryptocurrency market, *Chaos: Interdiscip. J. Nonlinear Sci.* 32 (8) (2022), 083142, <https://doi.org/10.1063/5.0104707>.
- [30] E.B. Wilson, J. Worcester, The determination of $\text{Id } 50$ and its sampling error in bio-assay, *Proc. Natl. Acad. Sci. U.S.A.* 29 (2) (1943) 79, <https://doi.org/10.1073/pnas.29.2.79>.
- [31] A.T. Azar, S.A. El-Said, Probabilistic neural network for breast cancer classification, *Neural Comput. Appl.* 23 (6) (2013) 1737–1751, <https://doi.org/10.1007/s00521-012-1134-8>.
- [32] G.W. Brier, et al., Verification of forecasts expressed in terms of probability, *Mon. Weather Rev.* 78 (1) (1950) 1–3, [https://doi.org/10.1175/1520-0493\(1950\)078<0001:VOFEIT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2).